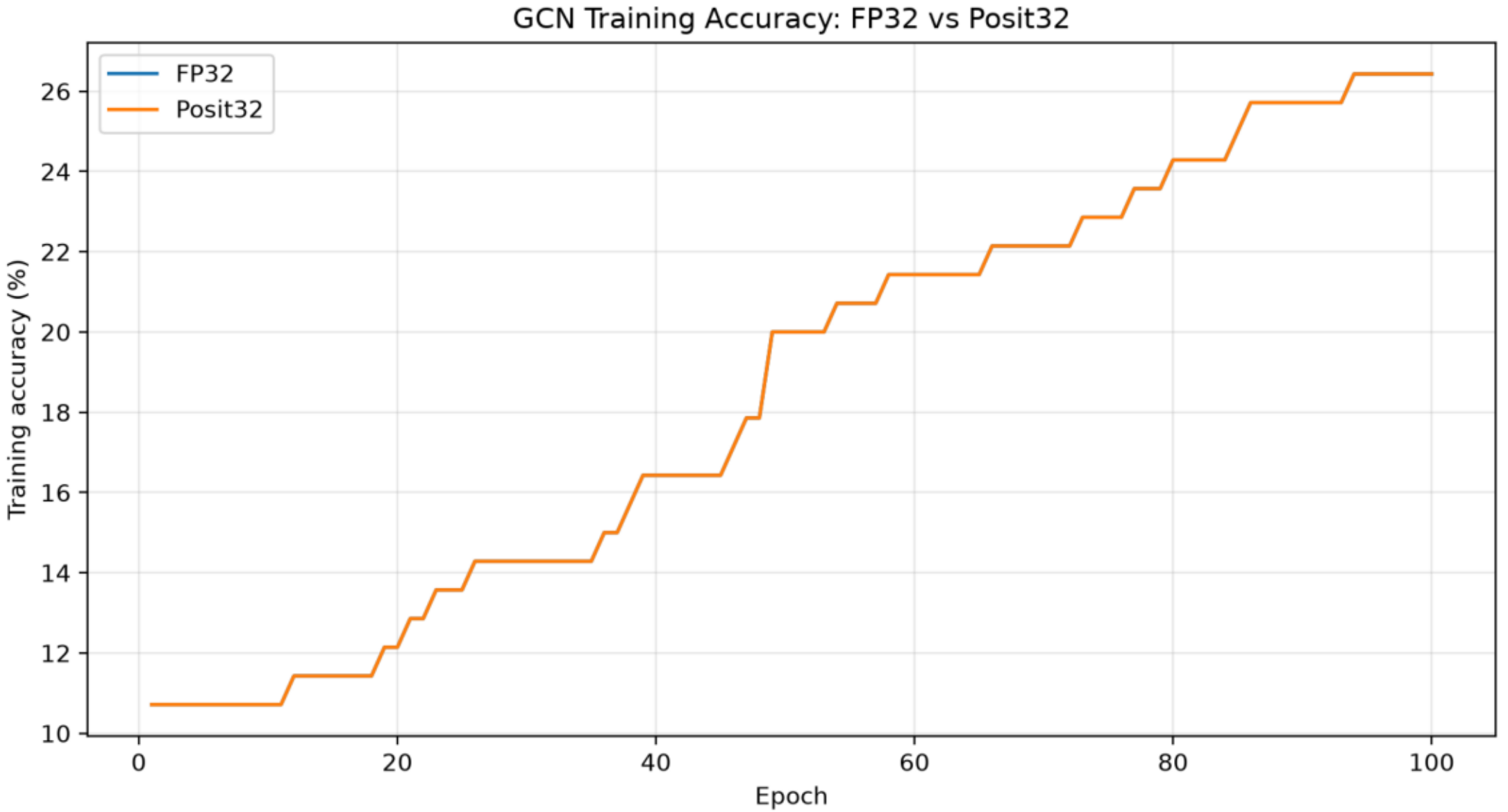
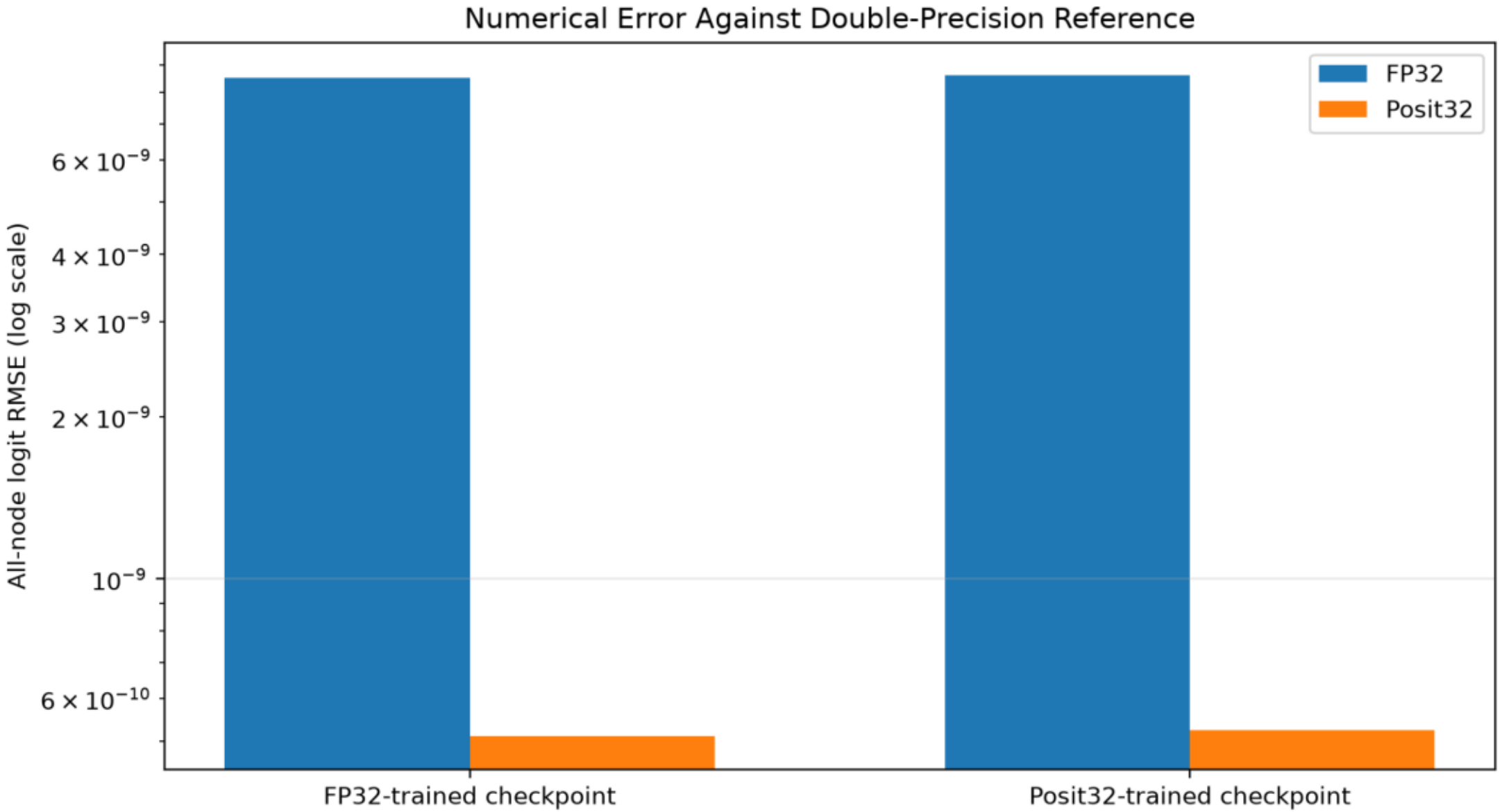
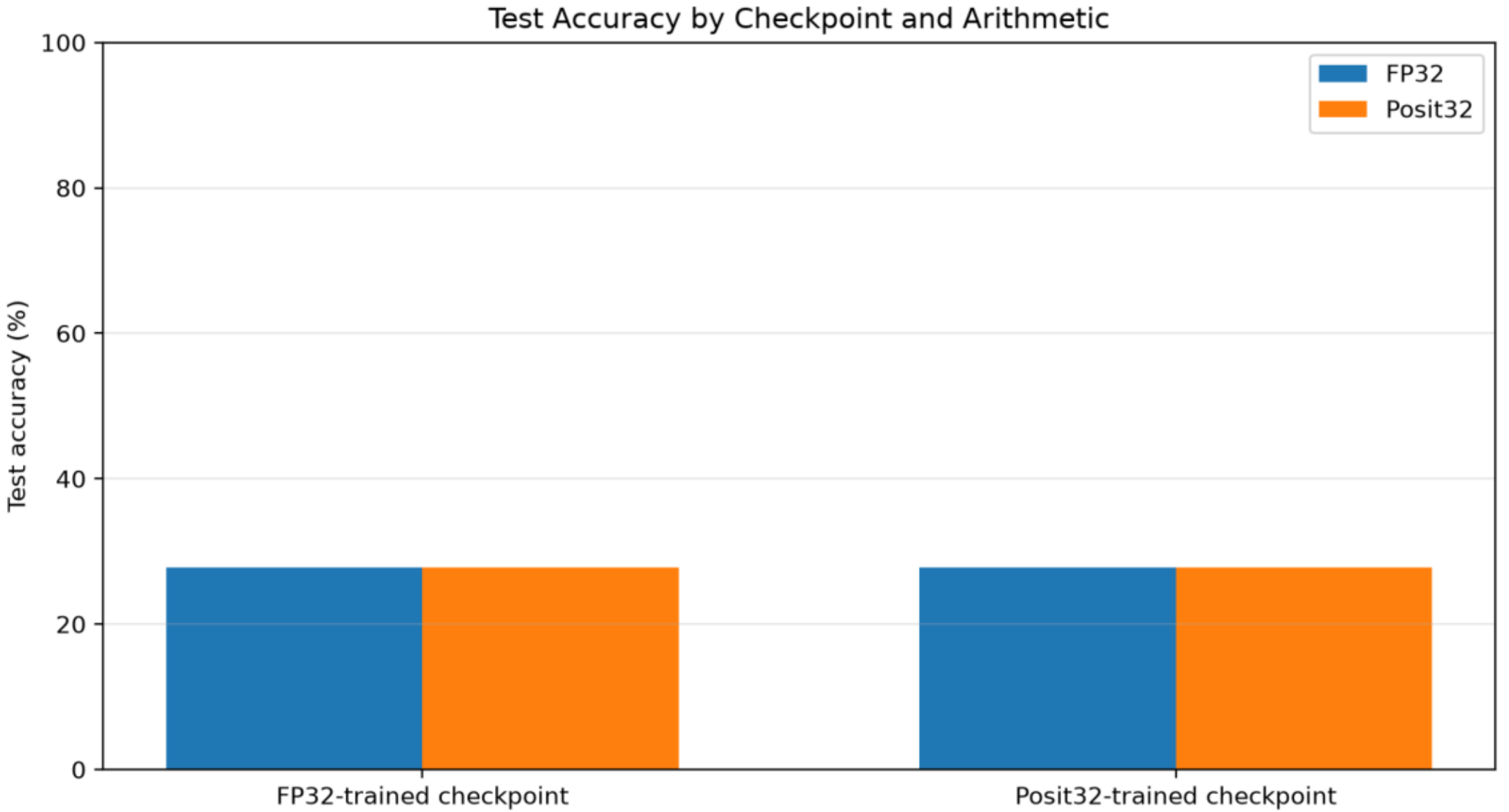

GCN Training Loss: FP32 vs Posit32
FP32
Posit32
Training loss
1.94
1.93
1.92
1.91
1.90
1.89
0
20
40
60
80
100
Epoch


GCN Training Accuracy: FP32 vs Posit32
FP32
Posit32
Training accuracy (%)
10
12
14
16
18
20
22
24
26
0
20
40
60
80
100
Epoch


Numerical Error Against Double-Precision Reference
FP32
Posit32
All-node logit RMSE (log scale)
6×10^{-9}
4×10^{-9}
3×10^{-9}
2×10^{-9}
10^{-9}
6×10^{-10}
FP32-trained checkpoint
Posit32-trained checkpoint


Test Accuracy by Checkpoint and Arithmetic
FP32
Posit32
Test accuracy (%)
100
80
60
40
20
0
FP32-trained checkpoint
Posit32-trained checkpoint